# A Contradictory Taste of Competitive Mathematics (😋∧🤔)

## CPMSoc Mathematics

Written By Brendan Mabbutt and Kevin Du

# Attendance

Please scan the QR code; and let us know about any dietary requirements.

(For the pizza!)

# What is Competitive Mathematics?

# Why Competitive Mathematics at UNSW?

- Meet people (connections for employability!)
- Having fun
- Solving real-world problems
- Cash prizes up for grabs in competitions
- Gives you reason to learn about interesting part of mathermatics
- Good if you like to discuss Math while eating Pizza (🍕)

Mathematics is distilled thinking

— Poh Shen-Loh

# Simon Marais Mathematics Competition (SMMC)

At UNSW this is the main competition for undergraduates to get involved in!

- Happens in mid-October
- Sign up by August through the university
- Mathematics department posts information for signing up closer to the date
- More than $50,000 in individual prizes, and internship offers 💵

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<u>Structure</u>

- Broken up into two sessions, 3 hours each
- Each session has 4 questions, for a total of 8
- Each problem is worth 7 marks regardless of difficulty
- Can include an open problem – marks awarded for reasonable progress

# Problems

What are the problems like?

- Proof-based ✍️
- Solutions are much easier to explain than to invent
- Require unusual techniques, observations, or combinations
- Usually chosen to be aesthetically pleasing 😍

*An exercise is a question that tests the student's mastery of a narrowly focused technique … the path towards solution is always apparent. In contrast, a problem is a question that cannot be answered immediately. Problems are often open-ended, paradoxical, and sometimes unsolvable, and require investigation before one can come close to a solution. Problems and problem solving are at the heart of mathematics.*

— Paul Zeitz

# Subject areas

Four main (very broad) classifications:

- **A**lgebra,
- **C**ombinatorics,
- **G**eometry,
- **N**umber theory.

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- Analysis
- Calculus
- Geometry ⬢
- Trigonometry ▲
- Vectors
- Complex numbers
- Polynomials

- Functional equations
- Inequalities $\leq$
- Combinatorics & probability
- Number theory ($6 \sim 7$)
- Graph theory
- Group theory
- Game theory

# Learning Competitive Mathematics

# How to Learn Problem-solving

## Strategies

- Practice strategically
- Select good notation
- Modify the problem
- Work backwards
- Consider concrete examples
- Make conjectures
- Reduce to something familliar
- Read solutions (only after attempt!)

## Techniques

- Induction
- Invariants
- Cases / exhaustion
- Pigeonhole principle
- Extremal principle
- Telescoping
- Symmetry
- Generating functions

# Resources

## Books

- Problem Solving Tactics (Pasquale et. al., 2014)
- The Art and Craft of Problem Solving (Zeitz, 2007)
- Putnam and Beyond (Gelca & Andreescu, 2007)
- Solving Mathematical Problems (Tao, 2006)

## Other

- YouTube
- The Putnam Archive
- Wikipedia (for theorems)
- Mathematical Olympiad Discord Server

# Proof by Contradiction

# General Form

**Definition**

*Proof by Contradiction is the following proof process:*

1. *We want to disprove* $P$.
2. *Suppose* $P$ *is true.* $(P \equiv \top)$
3. *Prove* $Q$ *follows from* $P$. $(P \Rightarrow Q)$
4. *Prove* $Q$ *is false.* $(Q \equiv \bot)$
5. *Then,* $\top \Rightarrow \bot$, *which cannot be, so* $P$ *is false.*

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**Law of Contraposition**

$$P \to Q \Leftrightarrow \neg Q \to \neg P$$

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## Example

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- Define $q = p_1 \times p_2 \times \ldots \times p_n + 1$. Then $q$ is larger than any prime, so it must be composite.

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- For each $p_i$, division of $q$ by $p_i$ leaves remainder 1. Hence $q$ is not divisible by any prime.
- Every composite number is divisible by a prime. This is a contradiction, so there must be infinitely many primes.

# Example Product Question

**Example**

*Let* $a_1, a_2, \ldots, a_n$ *be a rearrangement of the numbers* $1, 2, \ldots, n$*. Show that if* $n$ *is odd then* $A = (a_1 - 1)(a_2 - 2) \times \ldots \times (a_n - n)$ *is even.*

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- This means $a_1, a_3, a_5, \ldots, a_n$ are even, and $a_2, a_4, a_6, \ldots, a_{\{n-1\}}$ are odd.

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- So $a_1, a_2, \ldots, a_n$ contains $\frac{n+1}{2}$ even numbers and $\frac{n-1}{2}$ odd numbers.

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- So $a_1, a_2, \ldots, a_n$ contains $\frac{n+1}{2}$ even numbers and $\frac{n-1}{2}$ odd numbers.
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- But $1, 2, 3, \ldots, n$ contains $\frac{n-1}{2}$ even numbers and $\frac{n+1}{2}$ odd numbers.
- This is a contradiction, so $A$ must be even.

# Example Overlap Question 🎂 🍰

**Example**

*Brandon decorates the top of a cake with* $9$ *connected flat patches, that can be though of as planar surfaces*

*The union of the patches, each of area equal to* $1$*, has a total area equal to* $5$*. Prove that the overlap of some two of these patches has an area greater than or equal to* $\frac{1}{9}$*.*

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- Suppose that the overlap of every two surfaces is less than $\frac{1}{9}$.
- Number the surfaces arbitrarily $S_1, S_2, \ldots, S_9$.
- Then $S_1 \cup S_2$ has area greater than $1 + \left(1 - \frac{1}{9}\right) = 1 + \frac{8}{9}$.

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- Similarly, $S_1 \cup S_2 \cup S_3$ has area greater than $1 + \frac{8}{9} + \left(1 - \frac{1}{9} - \frac{1}{9}\right) = 1 + \frac{8}{9} + \frac{7}{9}$.

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- Continuing this pattern, $S_1 \cup S_2 \cup \ldots \cup S_9$ has area greater than $1 + \frac{8}{9} + \frac{7}{9} + \ldots + \frac{1}{9} = 5$.

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- This is a contradiction, so the overlap of some two surfaces must be at least $\frac{1}{9}$.

# Example Polynomial Question

**Example**

*Let $p(x)$ be a polynomial with even degree and positive leading coefficient.*
*Show that if $p(x) \geq p''(x)$, then $p(x) \geq 0$ for all real $x$.*

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- Then $p(a) - p''(a) < 0$, which contradicts the assumption $p(x) - p''(x) \geq 0$.

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- Then $p(a) - p''(a) < 0$, which contradicts the assumption $p(x) - p''(x) \geq 0$.
- Hence, $p(x) \geq 0$ for all real $x$.

# Example Partition Question 🍝

**Example**

*Suppose that Miles is trying to split a string of spaghetti between him and his friend. His friend Thomas suggests that the spaghetti be cut so that, for any length $x$ measured from the start of the spaghetti, Paul receives $x + a$, where $a$ is his favourite number.*

*Show that this is not possible, no matter what Paul's favourite number is.*

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*Show that the interval* $[0, 1]$ *cannot be partitioned into two disjoint sets* $A$ *and* $B$ *such that* $B = A + a$ *for some real number* $a$.

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- This means $[a, 2a) \subseteq B$, and we can continue this pattern using induction.
- Specifically, if $[2ka, (2k+1)a) \subseteq A$, then $[(2k+1)a, (2k+2)a) \subseteq B$. If $b \in [(2k+2)a, (2k+3)a)$ were in $B$ as well, then $b - a \in [(2k+1)a, (2k+2)a)$ would be in $A$, contradicting that $A$ and $B$ are disjoint.

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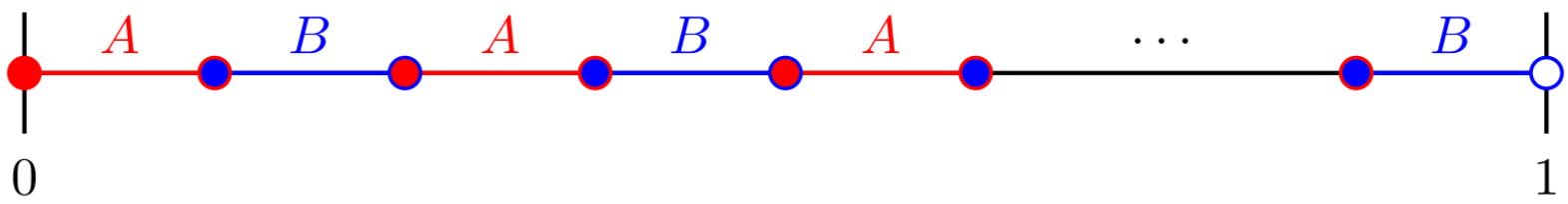

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- We know $A \cup B = [0, 1]$, so one of these intervals must contain 1. But then $1 \in [na, (n+1)a)$ for some natural number $n$, and this interval contains a number larger than 1.

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- We know $A \cup B = [0, 1]$, so one of these intervals must contain 1. But then $1 \in [na, (n+1)a)$ for some natural number $n$, and this interval contains a number larger than 1.
- Hence, it is impossible to partition $[0, 1]$ in this way.

# Example Irrational Question

**Example**

*Prove that* $\sqrt{2} + \sqrt{3} + \sqrt{5}$ *is an irrational number.*

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- Rearranging and squaring both sides, we get $5 + 2\sqrt{6} = r^2 + 5 - 2r\sqrt{5}$.

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- Then $\sqrt{6} + r\sqrt{5}$ must be rational.
- Squaring this gives $6 + 5r^2 + 2r\sqrt{30}$, which must be rational.

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- Then $\sqrt{6} + r\sqrt{5}$ must be rational.
- Squaring this gives $6 + 5r^2 + 2r\sqrt{30}$, which must be rational.
- This implies that $\sqrt{30}$ must be rational.

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*Prove that $\sqrt{2} + \sqrt{3} + \sqrt{5}$ is an irrational number.*

- Let $\sqrt{30} = \frac{p}{q}$ with $\gcd(p, q) = 1$ and $q \neq 0$.

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- Let $\sqrt{30} = \frac{p}{q}$ with $\gcd(p, q) = 1$ and $q \neq 0$.
- Squaring and rearranging gives $p^2 = 30q^2$. Then $p^2$ is even, so $p$ is even. Hence $p^2$ has a factor of 4.

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- Squaring and rearranging gives $p^2 = 30q^2$. Then $p^2$ is even, so $p$ is even. Hence $p^2$ has a factor of 4.
- Since $q^2 = \frac{p^2}{30}$, $q^2$ has a factor of 2, so $q$ is even. This contradicts $\gcd(p, q) = 1$.

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- Squaring and rearranging gives $p^2 = 30q^2$. Then $p^2$ is even, so $p$ is even. Hence $p^2$ has a factor of 4.
- Since $q^2 = \frac{p^2}{30}$, $q^2$ has a factor of 2, so $q$ is even. This contradicts $\gcd(p, q) = 1$.
- Therefore, $\sqrt{2} + \sqrt{3} + \sqrt{5}$ is irrational.

# Example Edible 5-Tuples Question 🍽️🥖

**Example**

*Suppose that Henry is setting up a combinatorics ball. He has a circular table with* $2026$ *seats. Henry loves the number* $29$*, and he wants the alternating sum of each group of* $5$ *consecutive plates to be equal to* $29$*, where the numbers represent the number of bread rolls he places on each plate. Determine all the ways Henry can arrange the numbers on the plates so that this condition is satisfied for every* $5$*-tuple of consecutive plates.*

# Example Edible 5-Tuples Question 🍽️🥖

**Example**

*We call a 5-tuple of integers edible if its elements can be labelled $a, b, c, d, e$ in some order so that $a - b + c - d + e = 29$. Determine all 2026-tuples of integers $n_1, n_2, \ldots, n_{2026}$ such that if placed in a circle clockwise, every 5-tuple of consecutive numbers is edible.*

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- First, subtract 29 from each $n_i$ so that the condition becomes $a - b + c - d + e = 0$.
- Consider the sum of five consecutive numbers:

$$\begin{aligned} n_i + n_{i+1} + n_{i+2} + n_{i+3} + n_{i+4} &= (n_i - n_{i+1} + n_{i+2} - n_{i+3} + n_{i+4}) + 2(n_{i+1} + n_{i+3}) \\ &= 2(n_{i+1} + n_{i+3}). \end{aligned}$$

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- Notice that this sum is even.

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- So $n_i + n_{i+1} + n_{i+2} + n_{i+3} + n_{i+4} - (n_{i+1} + n_{i+2} + n_{i+3} + n_{i+4} + n_{i+5})$ is even.

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- So $n_i - n_{i+5}$ is even, so $n_i$ and $n_{i+5}$ have the same parity.
- Then $n_1, n_6, \ldots, n_{2026}$ all have the same parity. Following the clockwise arrangement, this pattern continues until every $n_i$ has the same parity.

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- So $n_i - n_{i+5}$ is even, so $n_i$ and $n_{i+5}$ have the same parity.
- Then $n_1, n_6, \ldots, n_{2026}$ all have the same parity. Following the clockwise arrangement, this pattern continues until every $n_i$ has the same parity.
- Next, since $n_i + n_{i+1} + n_{i+2} + n_{i+3} + n_{i+4}$ is even, and all $n_i$ have the same parity, we conclude that every $n_i$ is even.

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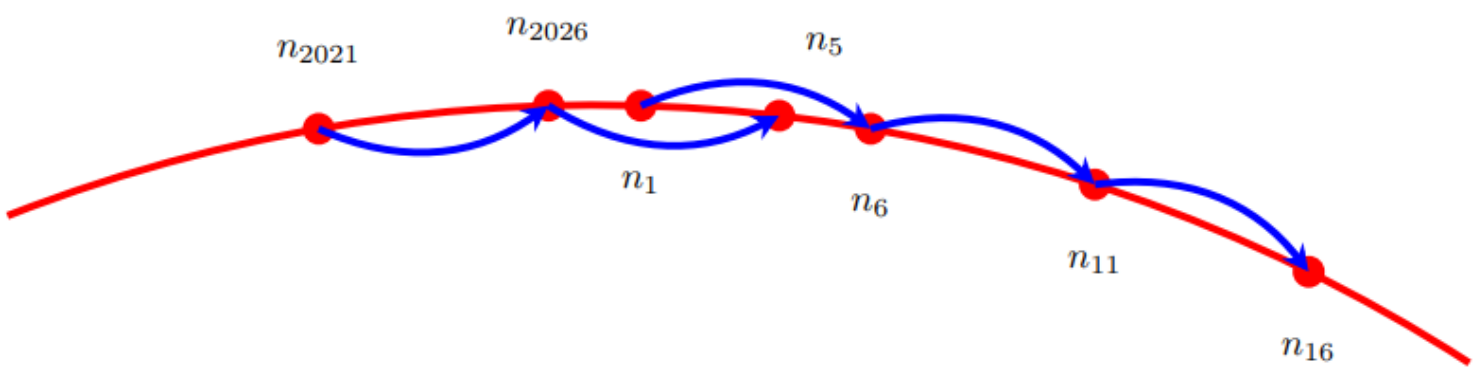

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- Now, consider a solution for which $\sum_{i=1}^{2026} |n_i| = S$ is minimal.
- Since every $n_i$ is even, we can divide each $n_i$ by 2 to get a new solution. If $S > 0$, the new solution has sum $\frac{S}{2} < S$, contradicting minimality.

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- Since every $n_i$ is even, we can divide each $n_i$ by 2 to get a new solution. If $S > 0$, the new solution has sum $\frac{S}{2} < S$, contradicting minimality.
- Hence, $S = 0$, which implies $n_i = 0$ for all $i$.
- Recall that we originally subtracted 29 from each $n_i$. Therefore, the only solution is $n_i = 29$ for all $i$.

# The End.

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